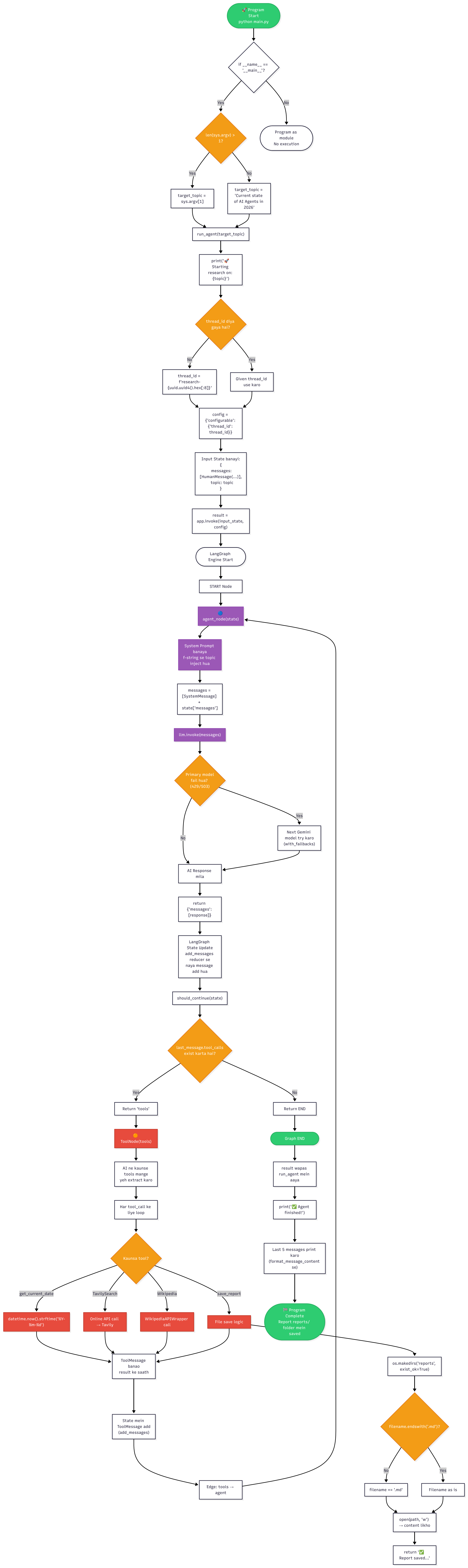
🚀 Program Start python main.py
if __name__ == '__main__'?
Yes
No
Program as module No execution
len(sys.argv) > 1?
Yes
No
target_topic = sys.argv[1]
target_topic = 'Current state of AI Agents in 2026'
run_agent(target_topic)
print('🚀 Starting research on: {topic}')
thread_id diya gaya hai?
No
Yes
thread_id = f'research-{uuid.uuid4().hex[:8]}'
Given thread_id use karo
config = {'configurable': {'thread_id': thread_id}}
Input State banayi: { messages: [HumanMessage(...)], topic: topic }
result = app.invoke(input_state, config)
LangGraph Engine Start
START Node
🔵 agent_node(state)
System Prompt banaya f-string se topic inject hua
messages = [SystemMessage] + state['messages']
llm.invoke(messages)
Primary model fail hua? (429/503)
Yes
No
Next Gemini model try karo (with_fallbacks)
AI Response mila
return {'messages': [response]}
LangGraph State Update add_messages reducer se naya message add hua
should_continue(state)
last_message.tool_calls exist karta hai?
Yes
No
Return 'tools'
Return END
🟠 ToolNode(tools)
Graph END
AI ne kaunse tools mange yeh extract karo
result wapas run_agent mein aaya
Har tool_call ke liye loop
print('✅ Agent finished!')
Last 5 messages print karo (format_message_content se)
Kaunsa tool?
get_current_date
TavilySearch
Wikipedia
save_report
datetime.now().strftime('%Y-%m-%d')
Online API call → Tavily
WikipediaAPIWrapper call
File save logic
🏁 Program Complete Report reports/ folder mein saved
ToolMessage banao result ke saath
State mein ToolMessage add (add_messages)
Edge: tools → agent
os.makedirs('reports', exist_ok=True)
filename.endswith('.md')?
No
Yes
filename += '.md'
Filename as is
open(path, 'w') → content likho
return '✅ Report saved...'